AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

1.Create an EC2 Windows Instance in AWS

Kuboja Daniel M

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

Coimbatore.

Objective

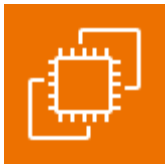
EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Windows Instances:

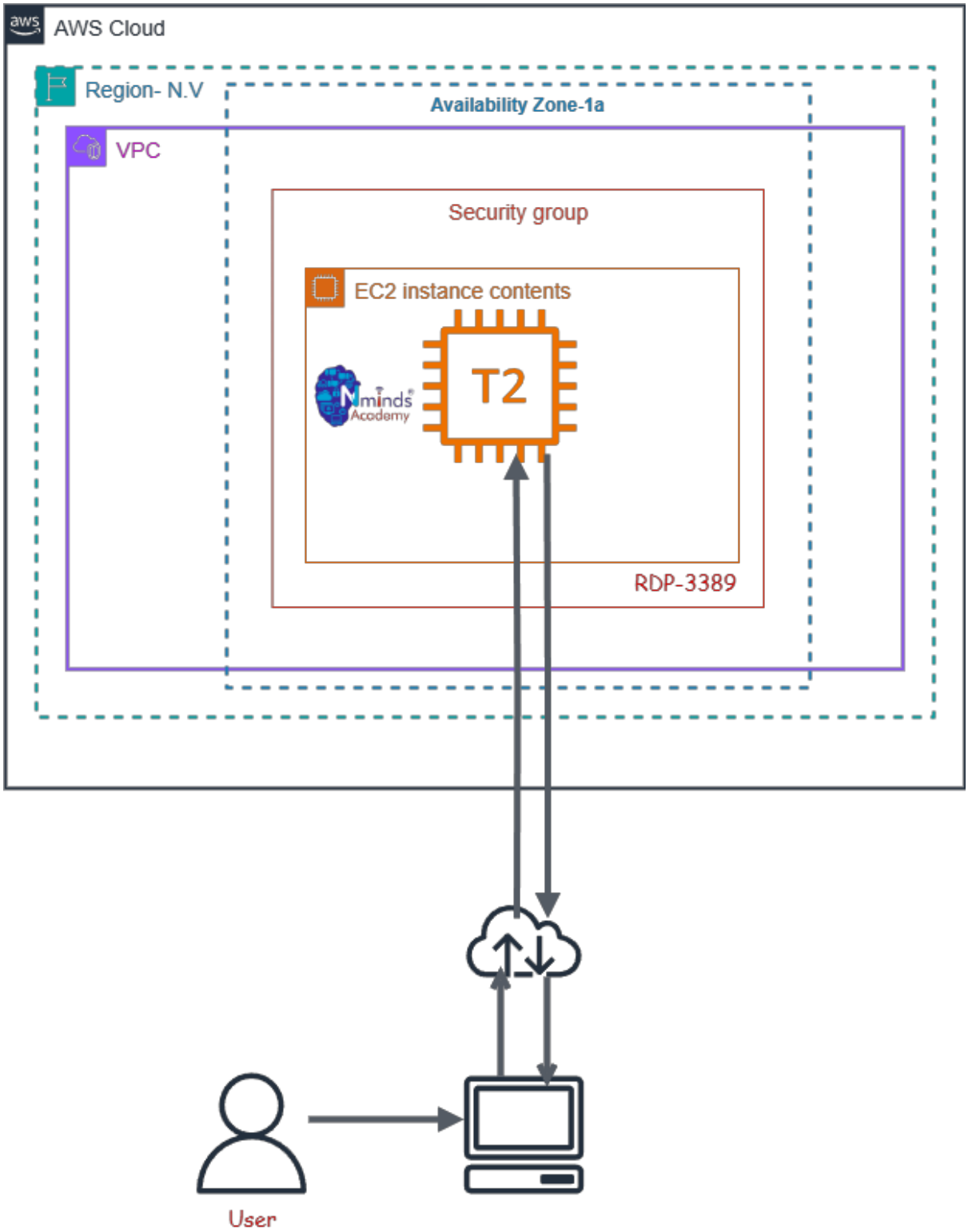
- Web and Application Hosting: Run IIS web servers or other Windows-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Windows instances, using RDS or EC2 for more control.
- Remote Desktop Applications: Set up RDS (Remote Desktop Services) for remote access to Windows desktops in the cloud.
- Development and Testing: Create development environments with Windows OS to test .NET or other Windows-based applications.

Used Services in AWS:

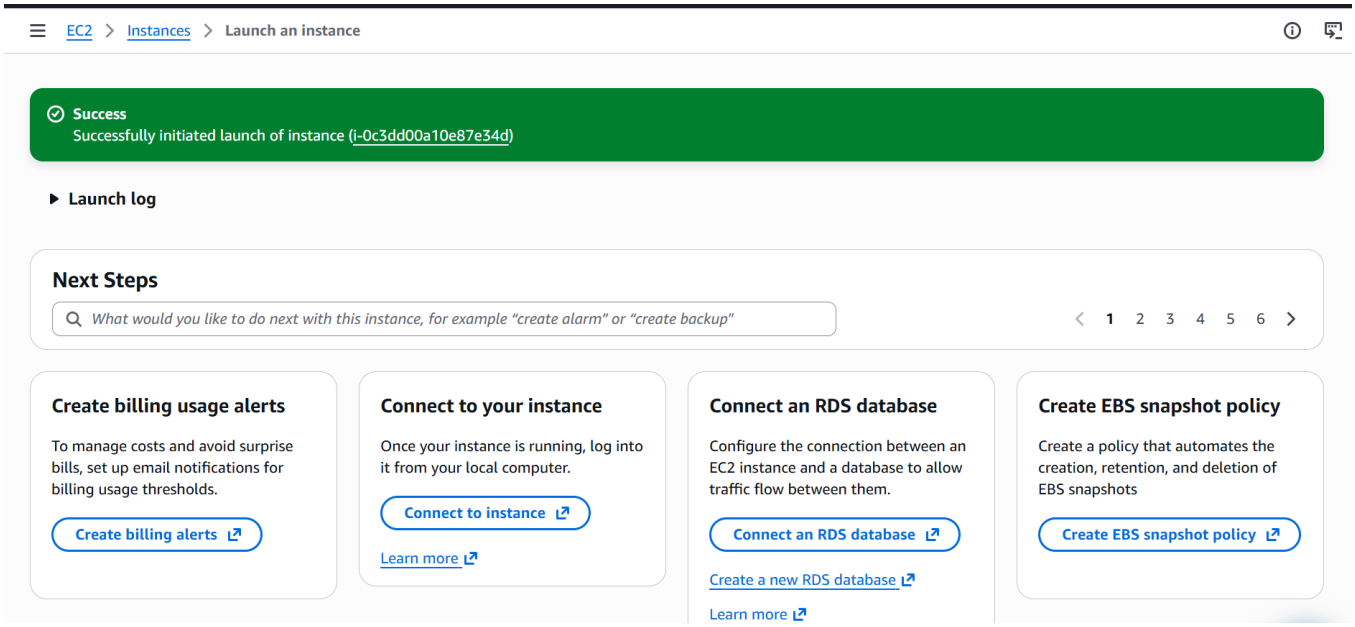


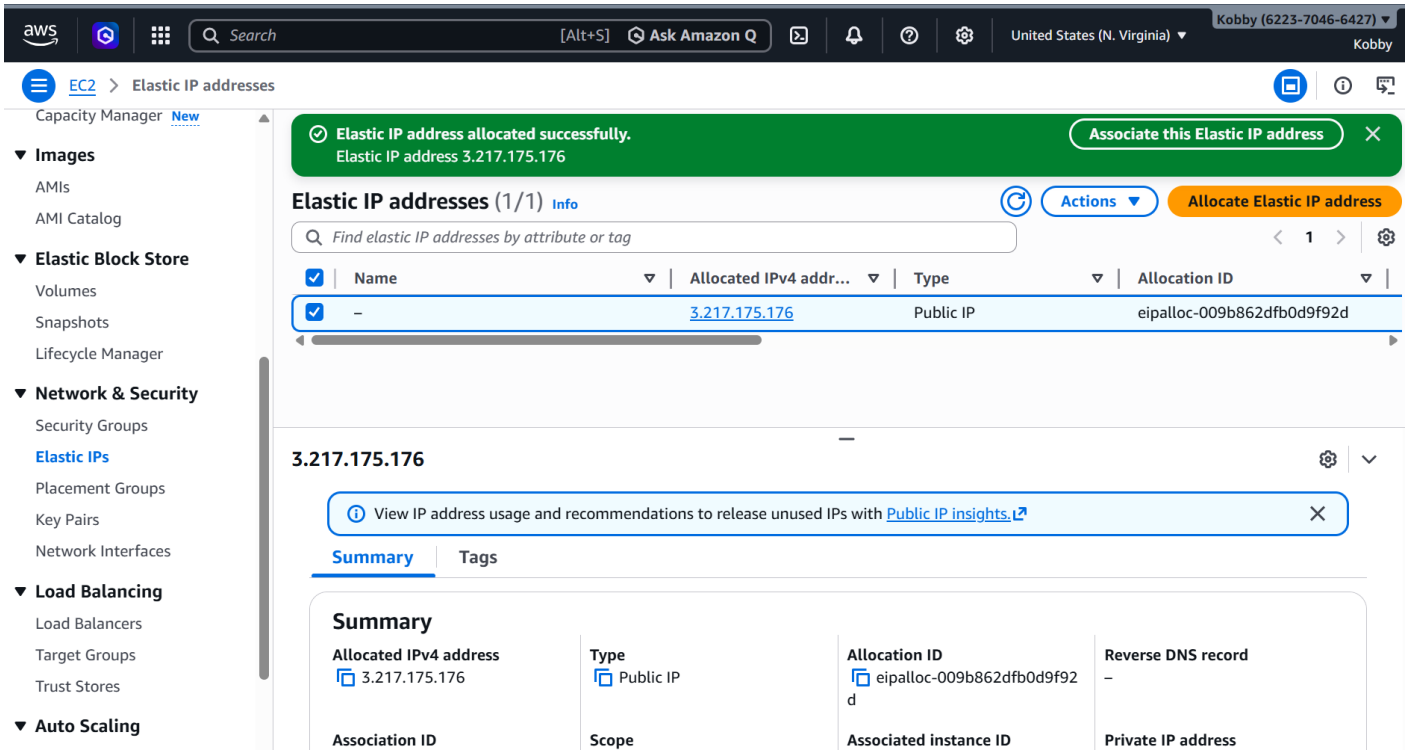
Amazon EC2

Topology


AWS Cloud
Region- N.V
Availability Zone-1a
VPC
Security group
EC2 instance contents
T2
RDP-3389
User

Execution Tasks:





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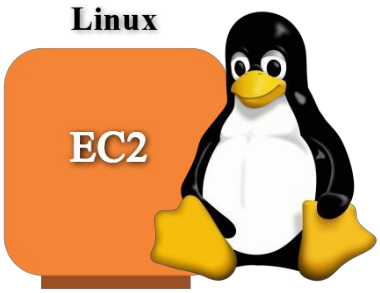
4.Create an EC2 Linux Instance in AWS



Aravindraj.G

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

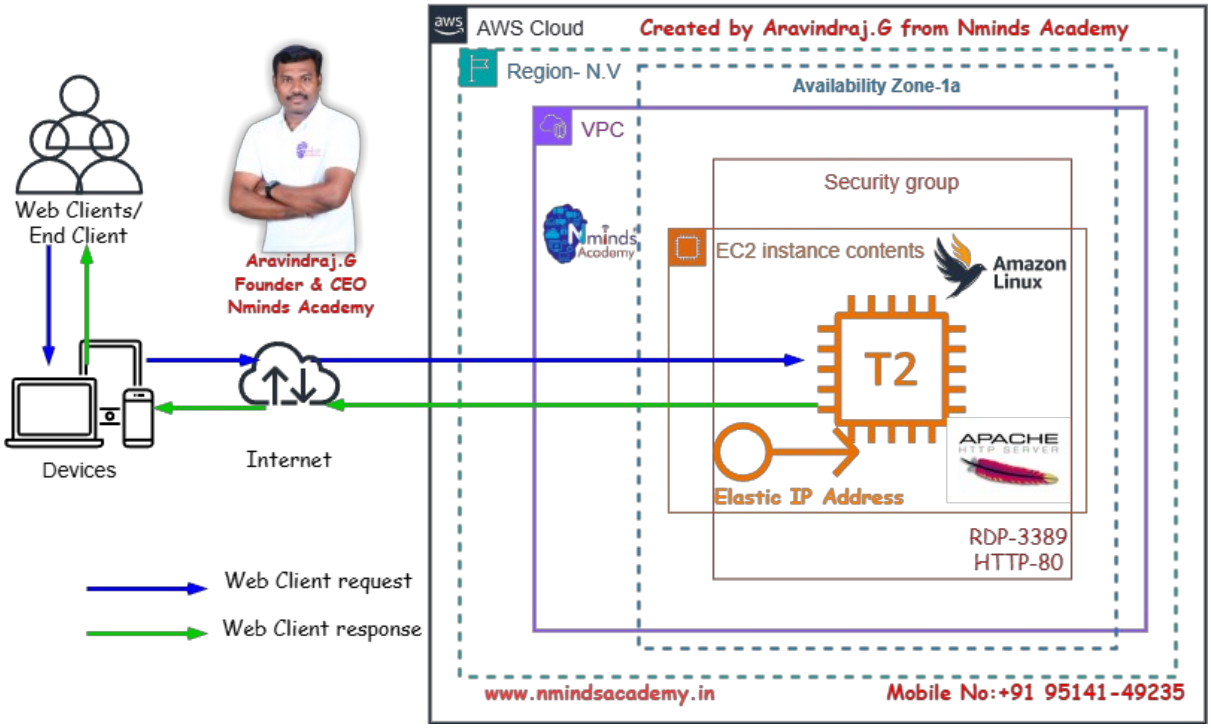
Execution Tasks:

```
[ec2-user@ip-172-31-15-3 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-03-02 15:22:31 UTC; 36min ago
       Docs: man:httpd.service(8)
   Main PID: 25514 (httpd)
     Status: "Total requests: 132; Idle/Busy workers 100/0;Requests/sec: 0.06; Bytes served/sec:  31 B/sec"
      Tasks: 230 (limit: 1067)
     Memory: 17.5M
        CPU: 2.371s
     CGroup: /system.slice/httpd.service
             ├─25514 /usr/sbin/httpd -DFOREGROUND
             ├─25515 /usr/sbin/httpd -DFOREGROUND
             ├─25516 /usr/sbin/httpd -DFOREGROUND
             ├─25529 /usr/sbin/httpd -DFOREGROUND
             ├─25545 /usr/sbin/httpd -DFOREGROUND
             └─26575 /usr/sbin/httpd -DFOREGROUND

Mar 02 15:22:31 ip-172-31-15-3.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 02 15:22:31 ip-172-31-15-3.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 02 15:22:31 ip-172-31-15-3.ec2.internal httpd[25514]: Server configured, listening on: port 80
[ec2-user@ip-172-31-15-3 ~]$ ls -l /var/www/html/
total 668
-rw-r--r--. 1 root     root       2091 Mar  2 15:46 index.html
-rw-rw-r--. 1 ec2-user ec2-user 678241 Mar  2 15:49 profile_bw.png
[ec2-user@ip-172-31-15-3 ~]$ curl localhost
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>About Me</title>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <style>
        body {
            margin: 0;
            font-family: Arial, Helvetica, sans-serif;
            background: linear-gradient(135deg, #1e3c72, #2a5298);
            display: flex;
            justify-content: center;
```

```
tcpdump:x:72:
screen:x:84:
ec2-user:x:1000:
apache:x:48:
kubo-dev:x:1001:
dan-dev:x:1002:
developer:x:1003:kubo-dev,dan-dev
[root@ip-172-31-15-3 ~]# passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-15-3 ~]# passwd ec2-user
Changing password for user ec2-user.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-15-3 ~]# passwd dan-dev
Changing password for user dan-dev.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-15-3 ~]# ls -ld /var/www/html
drwxr-xr-x. 2 root root 46 Mar  2 16:04 /var/www/html
[root@ip-172-31-15-3 ~]# chgrp developer1  /var/www/hmtl
chgrp: invalid group: ‘developer1’
[root@ip-172-31-15-3 ~]# chgrp developer /var/www/html
[root@ip-172-31-15-3 ~]# ls -ld /var/www/html
drwxr-xr-x. 2 root developer 46 Mar  2 16:04 /var/www/html
[root@ip-172-31-15-3 ~]# ls -ld /var/www/html
drwxr-xr-x. 2 root developer 46 Mar  2 16:04 /var/www/html
[root@ip-172-31-15-3 ~]# chmod 775 /var/www/html
[root@ip-172-31-15-3 ~]# vi /etc/ssh/sshd_config
[root@ip-172-31-15-3 ~]#  133L, 3855B written
[root@ip-172-31-15-3 ~]# systemctl reload sshd
[root@ip-172-31-15-3 ~]#
[root@ip-172-31-15-3 ~]# Connection to 3.238.93.235 closed by remote host.
Connection to 3.238.93.235 closed.

C:\Users\kubod\Downloads>
```

Topology

How to Configure the Elastic IP Address to Linux Web Server with AWS EC2



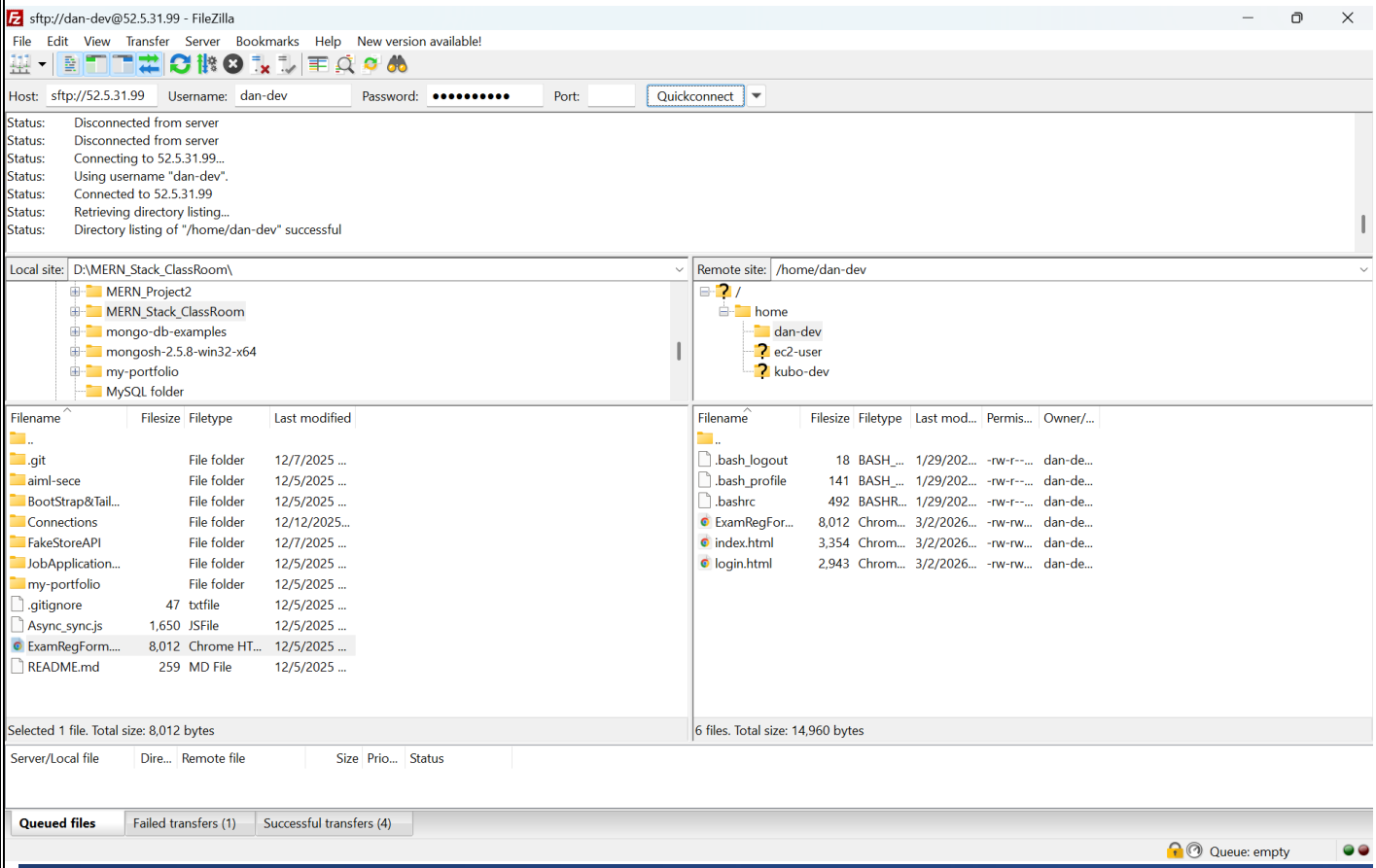
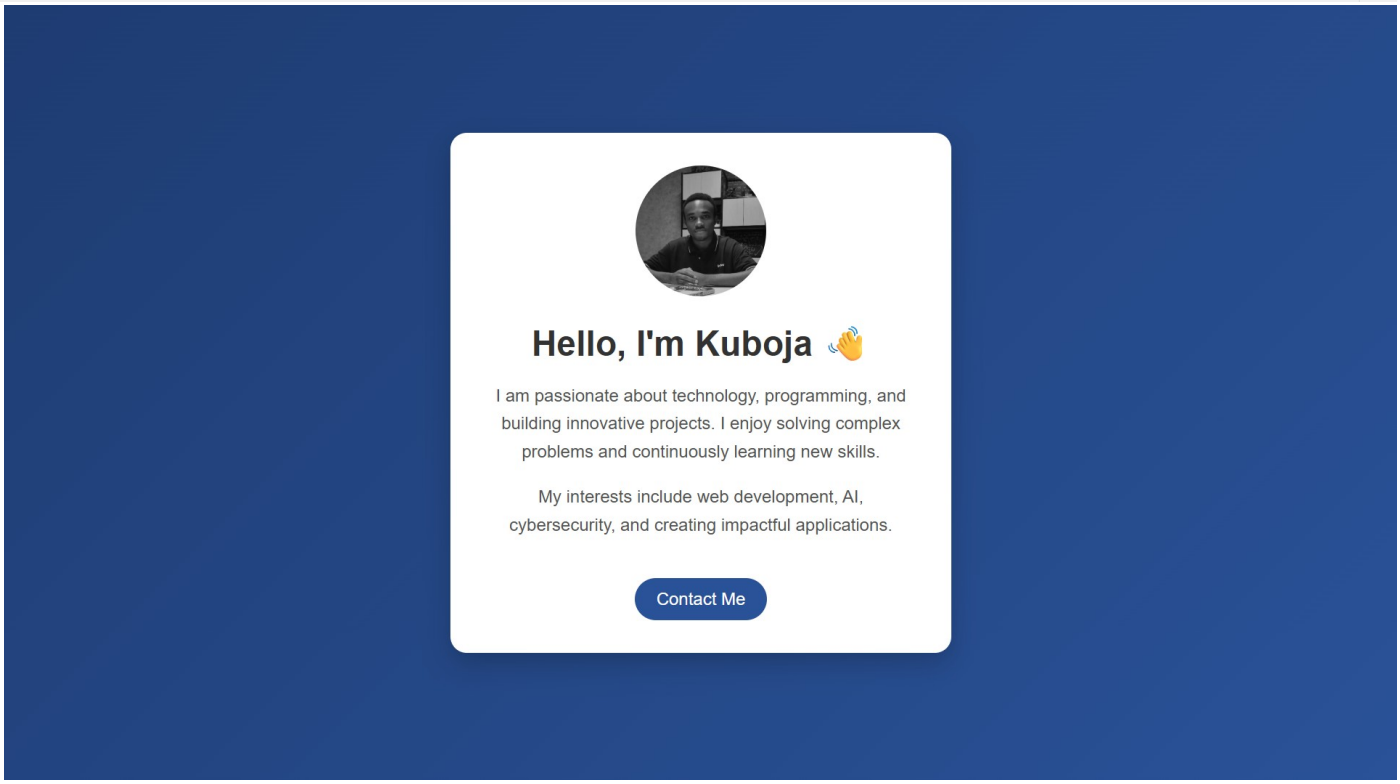
Used Services in AWS

Amazon EC2

Amazon Elastic IP Address

Used Services in Windows Server

APACHE HTTP SERVER

Apache Webserver

Execution Tasks: